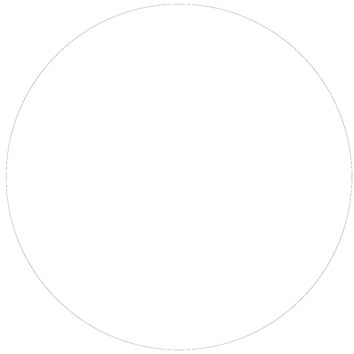
The Power of Rerandomization in Obfustopia

CRH, Somewhere-Extractable BARGs, and More...

Shafik Nassar

The University of Texas at Austin

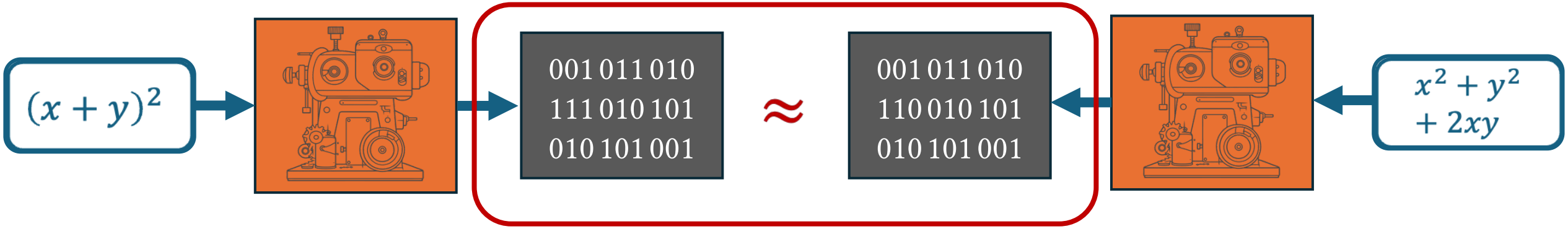


Indistinguishability Obfuscation[BGI+,GGH+]

- iO is a randomized algorithm: inputs a program, and outputs an obfuscated program



- Security: if $\forall x : P_0(x) = P_1(x)$, then $iO(P_0) \approx iO(P_1)$




iO and OWFs
can do anything!

iO + OWFs



Collision Resistant Hash

* black-box separation

Rerandomizable OWFs

Can be built from discrete log, factoring*...
or group actions

Not known to imply CRH!

One-way function:

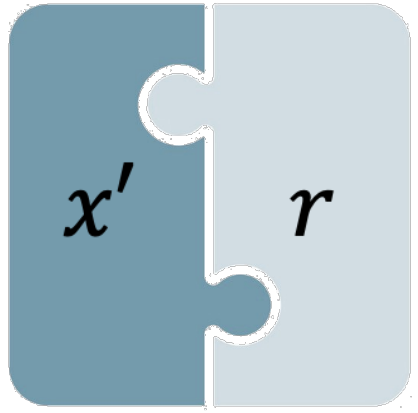
$$f(x) \to y$$

Rerandomize:

$$\mathrm{RR}(y; r) \to y'$$

Perfect rerandomization: $f(U_X) \equiv \mathrm{RR}(y; U_R)$

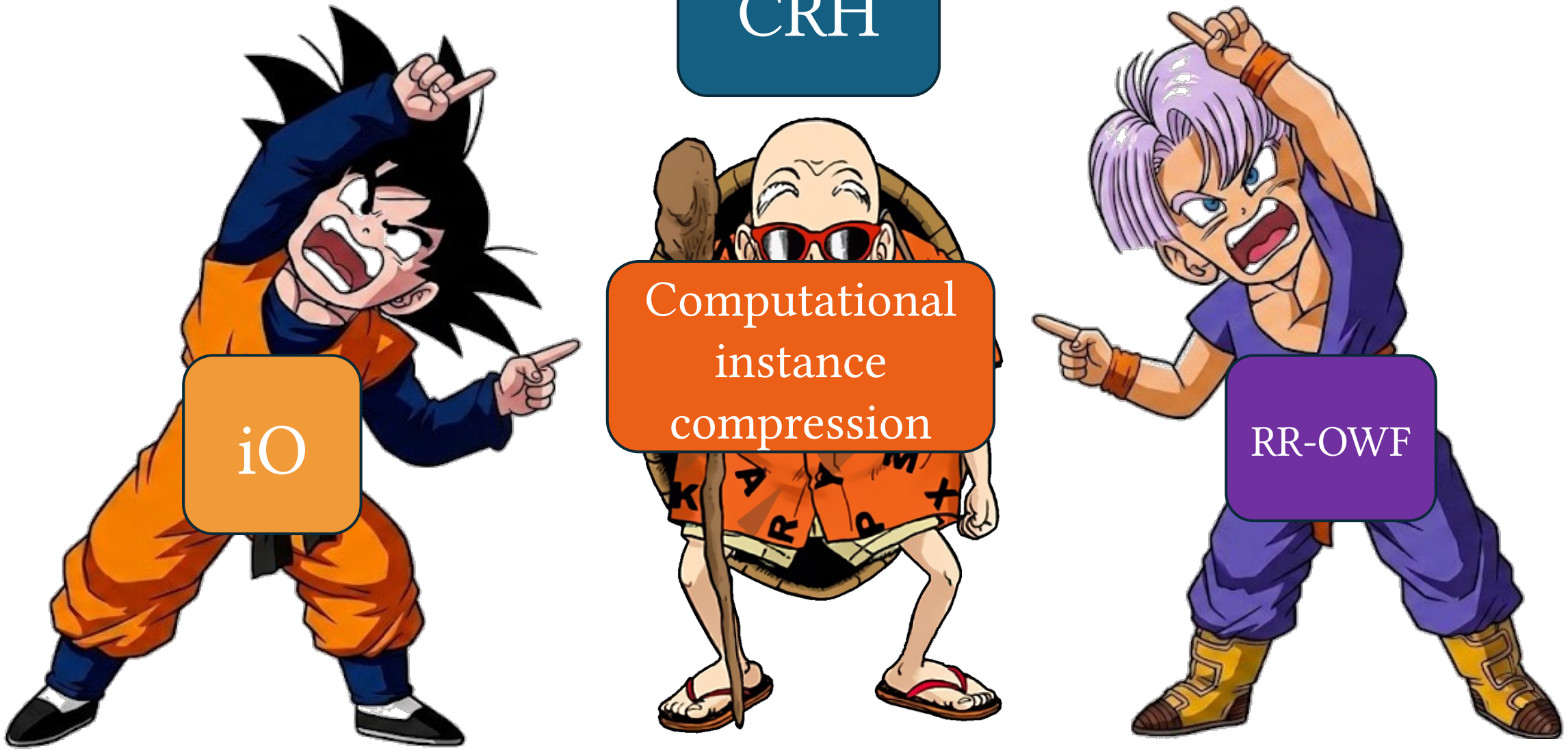
Preimage recovery: given r such that $\mathrm{RR}(y; r) = y'$ and preimage x' such that $f(x') = y'$ we can compute a preimage of y



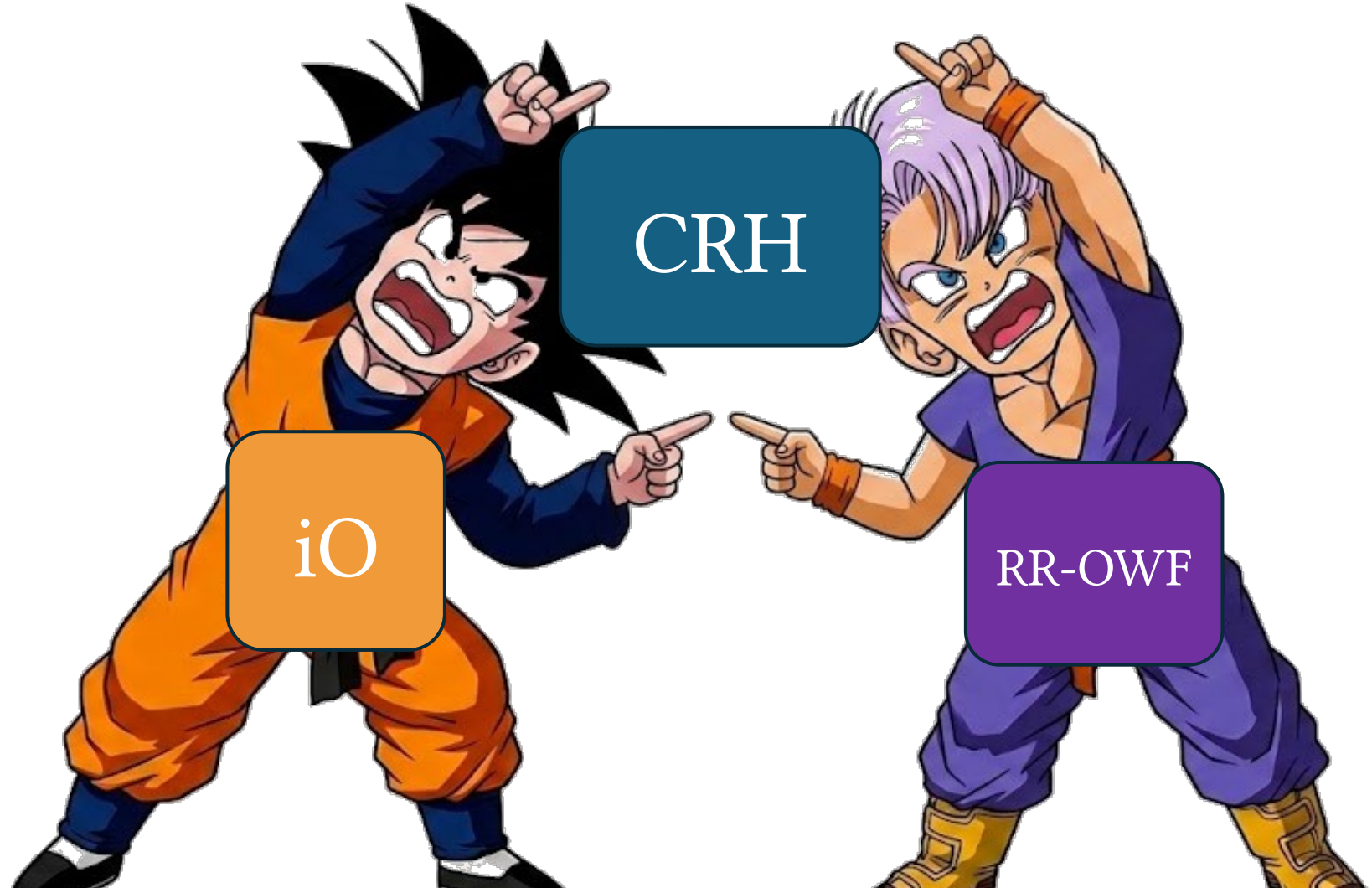
Question?



Prior Work [Arnon, Ben-David, Yogev 25]

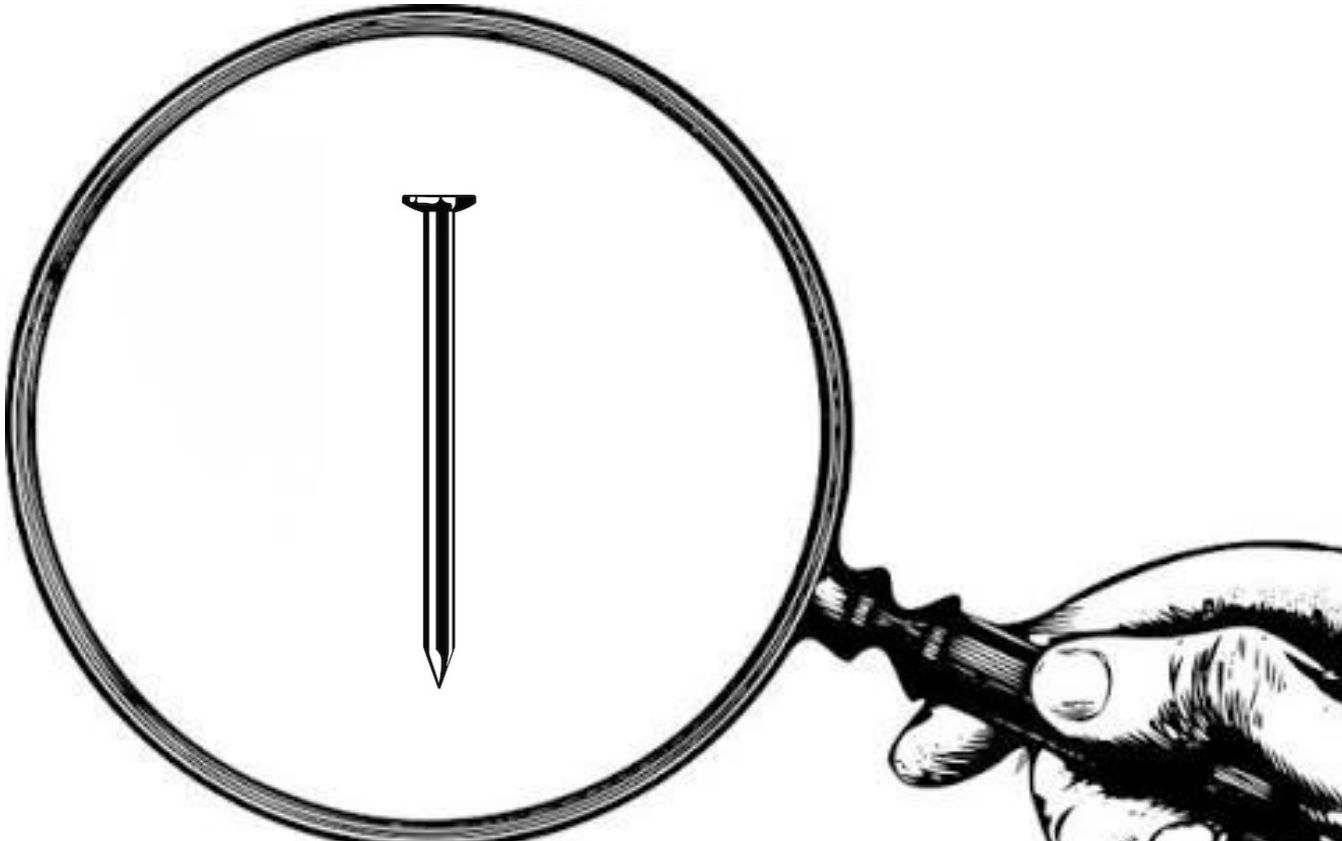


This Work: Result 1

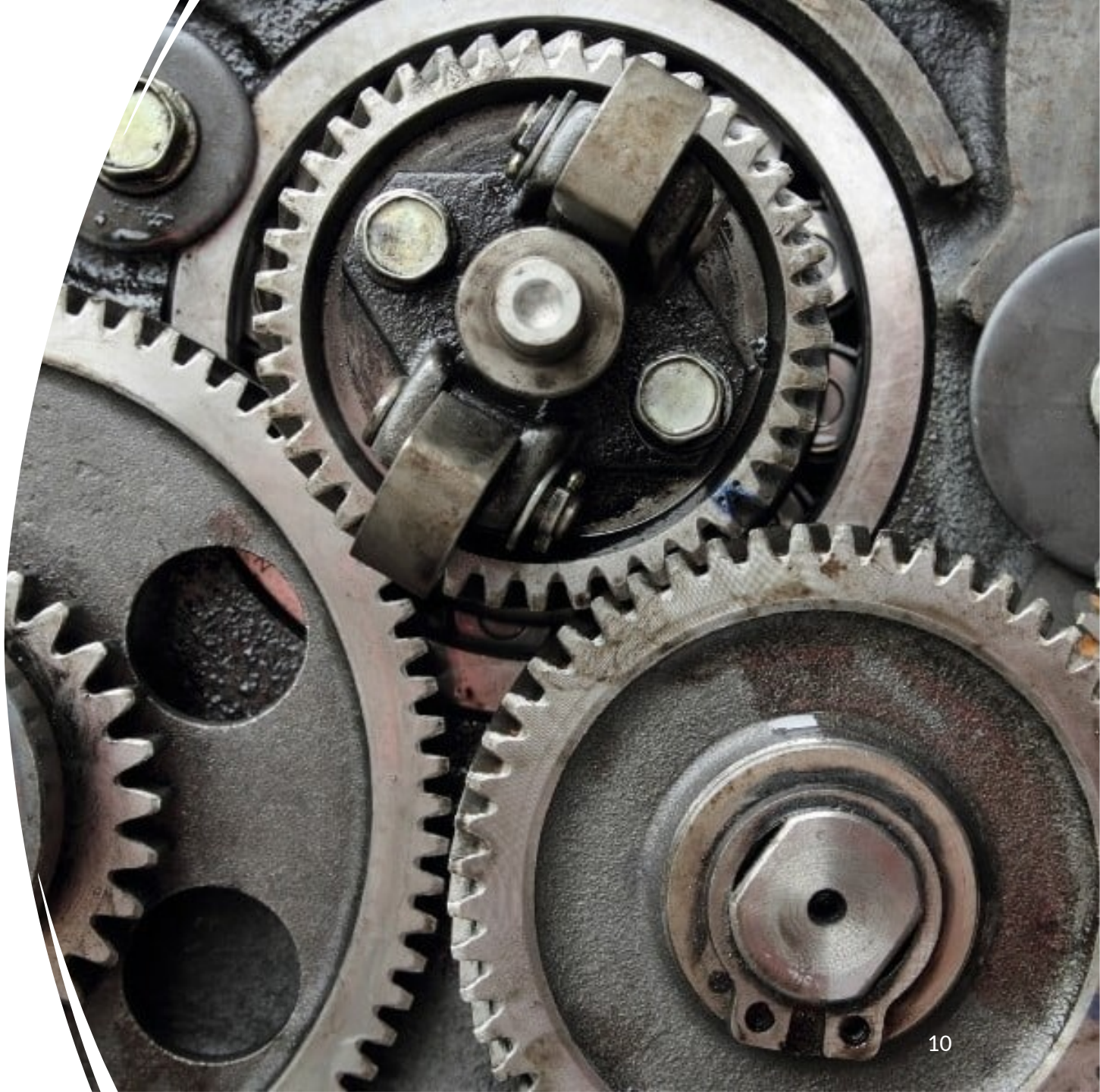




Applications to succinct arguments from same techniques



CRH Construction

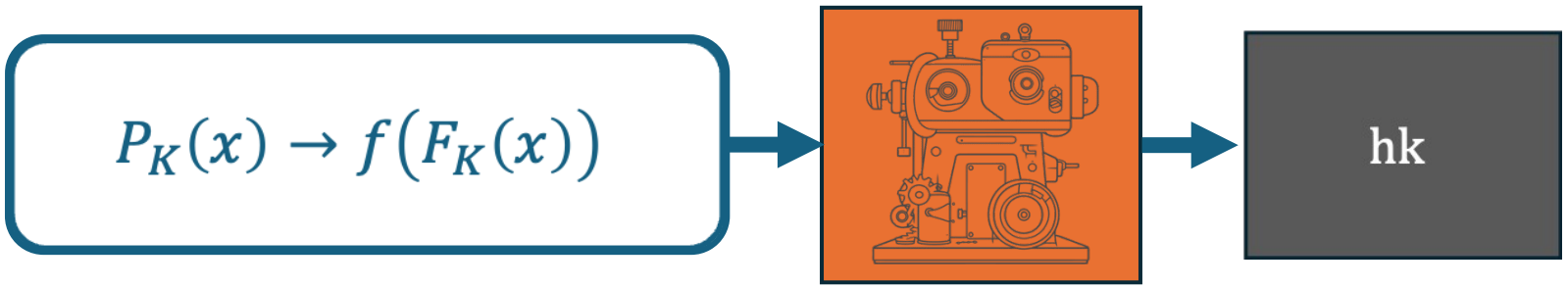


CRH from iO + PPRF + RR-OWF

Setup :

Rerandomizable OWF $f : \{0,1\}^n \to \{0,1\}^n$

Puncturable PRF $F_K : \{0,1\}^m \to \{0,1\}^n, m > n$

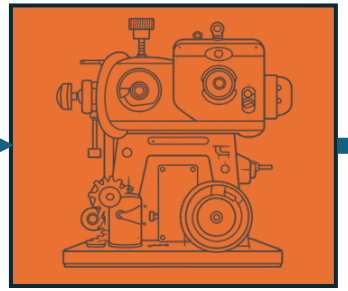


$\mathrm{Hash}(\mathrm{hk}, x) = \mathrm{hk}(x)$

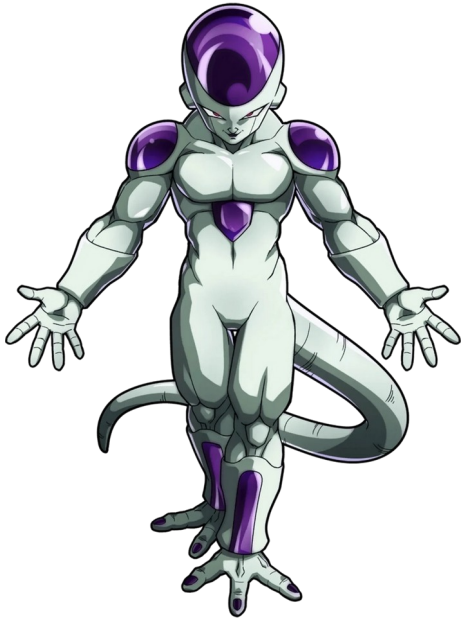
Security



$P_K(x) \rightarrow f\left(F_K(x)\right)$



hk



$x, x' \in \{0,1\}^m$

Adversary wins if

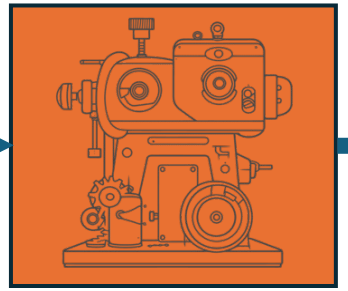
$x \neq x'$ and $\mathrm{hk}(x) = \mathrm{hk}(x')$

Security

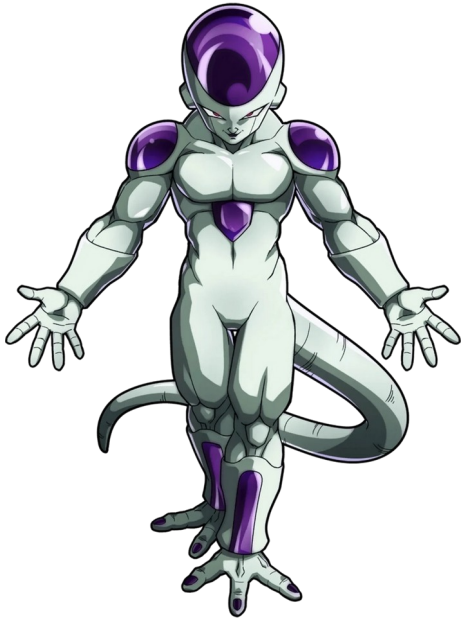


$i \leftarrow [m]$

$P_K(x) \rightarrow f\left(F_K(x)\right)$



hk



$x, x' \in \{0,1\}^m$

Adversary wins if

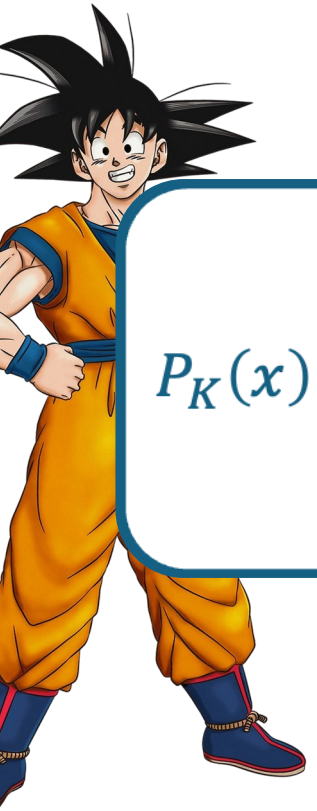
$x_i \neq x_i'$ and $\mathrm{hk}(x) = \mathrm{hk}(x')$

Security

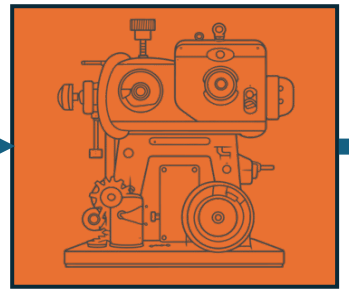
$i \leftarrow [m]$

Samples $\hat{x} \leftarrow \{0,1\}^m$

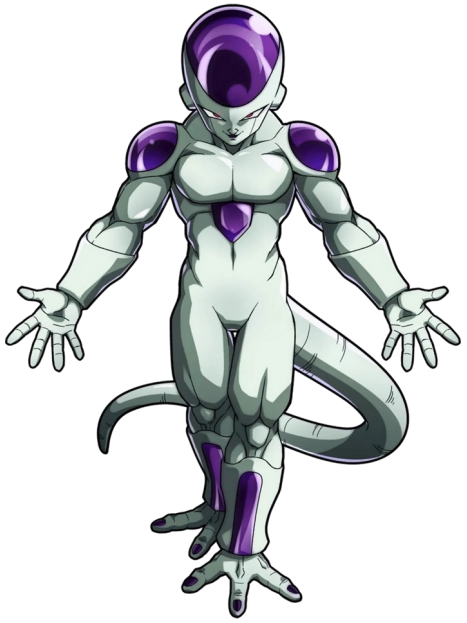
and sets $\hat{y} = f(\hat{x})$



$$P_K(x) \rightarrow \begin{cases} f(F_K(x)) & x_i = 0 \\ \mathrm{RR}(\hat{y}\,; F_K(x)) & x_i = 1 \end{cases}$$



hk



$x, x' \in \{0,1\}^m$

Adversary wins if

$x_i \neq x_i'$ and $\mathrm{hk}(x) = \mathrm{hk}(x')$

Security

$$P_K(x) \to f\left(F_K(x)\right) \quad \approx \quad P_K(x) \to \begin{cases} f\left(F_K(x)\right) & x_i = 0 \\ \mathrm{RR}\left(\hat{y}\,; F_K(x)\right) & x_i = 1 \end{cases}$$

2^m hybrids, change behavior one input at a time

Change behavior on x^*: puncture K on x^* iO security

1. Hardcode $y^* = f\left(F_K(x^*)\right)$

PPRF security

2. Sample $y^* \leftarrow f(U)$

Perfect rerandomization

3. Sample $y^* \leftarrow \mathrm{RR}(\hat{y}\,; U)$

PPRF security

4. Set $y^* = \mathrm{RR}\left(\hat{y}\,; F_K(x^*)\right)$

Security

Reduction

$$P_K(x) \rightarrow \begin{cases} f\left(F_K(x)\right) & x_i = 0 \\ \mathrm{RR}\left(\hat{y}\,; F_K(x)\right) & x_i = 1 \end{cases}$$

CRH Adversary

OWF challenge $\hat{y}$

Sends x, x'

$x_i = 0, x_i' = 1$

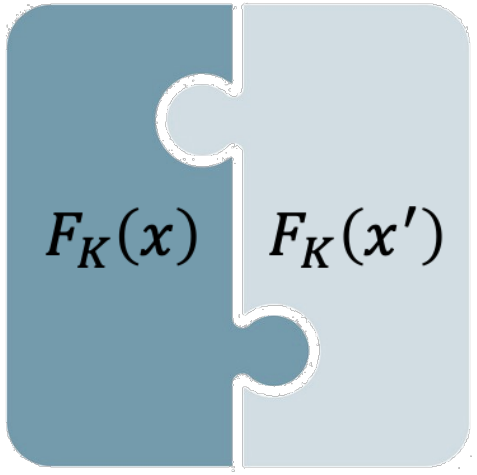
and $\mathrm{hk}(x) = \mathrm{hk}(x') = y$

$f\left(F_K(x)\right) = y$ preimage

$\mathrm{RR}\left(\hat{y}; F_K(x')\right) = y$ randomness

$f(\hat{x}) = \hat{y}$

$\hat{x} \leftarrow$



Direct CRH from iO+RR-OWF

Generalization

Perfectly
Partitionable
Hash

Use rerandomizable encryption

$\widehat{ct} \leftarrow \mathrm{Enc}(0) \quad \Rightarrow \quad \widehat{ct} \leftarrow \mathrm{Enc}(1)$

Normal hash key:

$P_K(x) \rightarrow \mathrm{Enc}(0\,; F_K(x))$

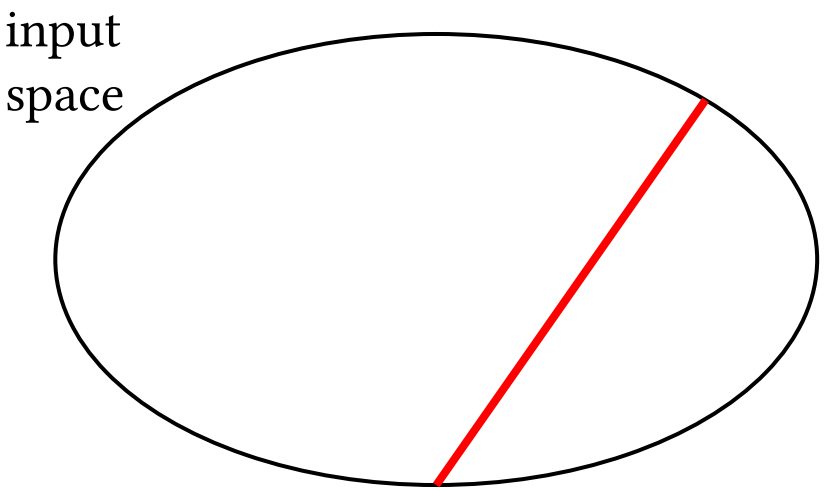
Partitioned hash key:

$$P_K(x) \rightarrow \begin{cases} \mathrm{Enc}(0; F_K(x)) & x_i = 0 \\ \mathrm{Enc}(1\,; F_K(x)) & x_i = 1 \end{cases}$$

Collisions between different partitions do NOT exist

Generalization

Perfectly
Partitionable
Hash

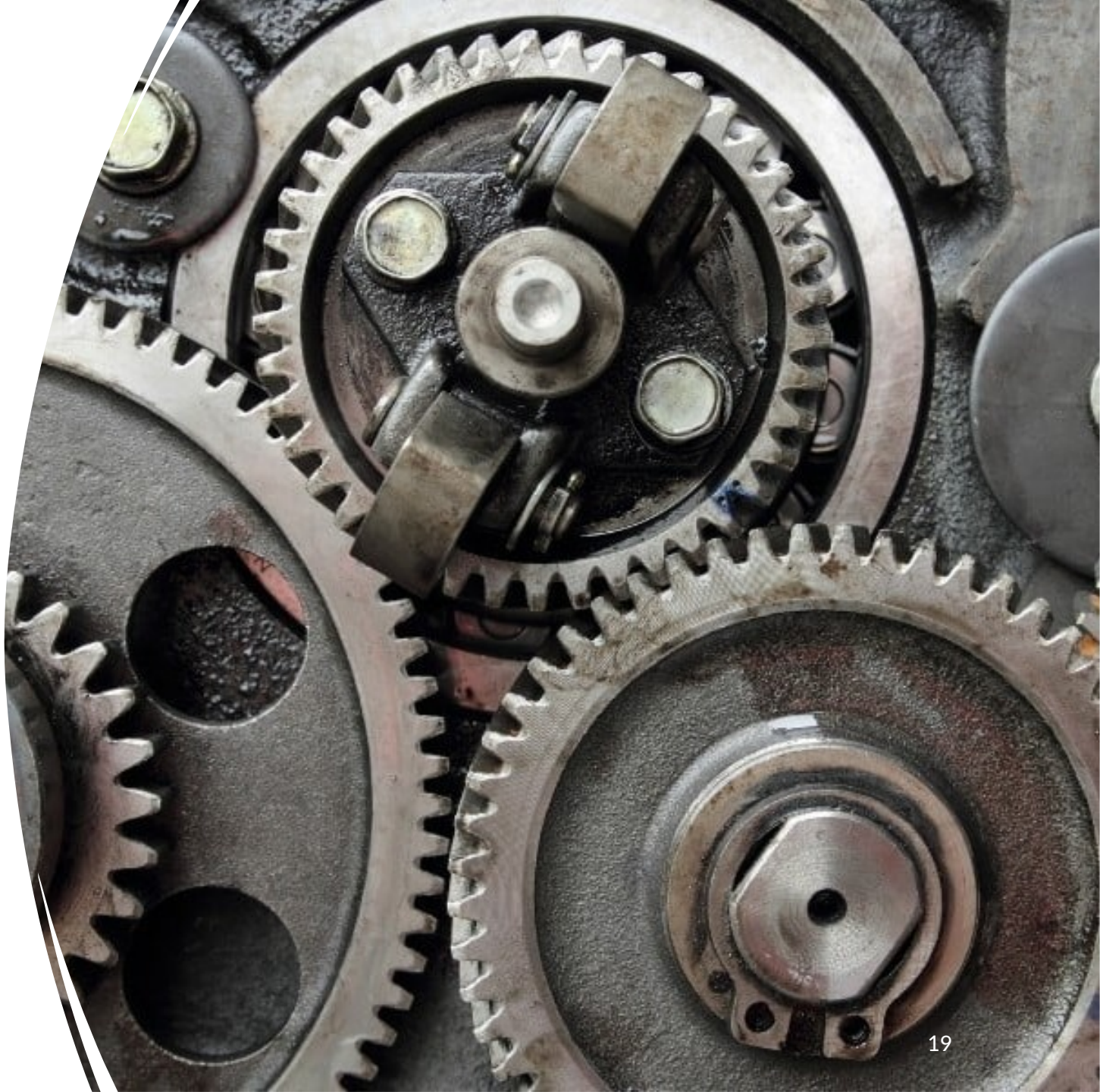


Partitioned hash key:

$$P_K(x) \rightarrow \begin{cases} \mathrm{Enc}(0; F_K(x)) & \mathrm{pred}(x) = 0 \\ \mathrm{Enc}(1; F_K(x)) & \mathrm{pred}(x) = 1 \end{cases}$$

Collisions between different partitions do NOT exist

Applications to Proof Systems



SNARG

$\mathrm{CRS} \leftarrow \mathrm{Setup}$



x

x, w



π

Complete: If $(x, w) \in \mathcal{R}$, verifier accepts

Sound: If $x \notin \mathcal{L}$, hard to find accepting π

Succinct: $|\pi| \ll |w|$

Sometimes-Statistically Sound SNARG

CRS ← Setup

CRS′ ← Setup(pred)



x

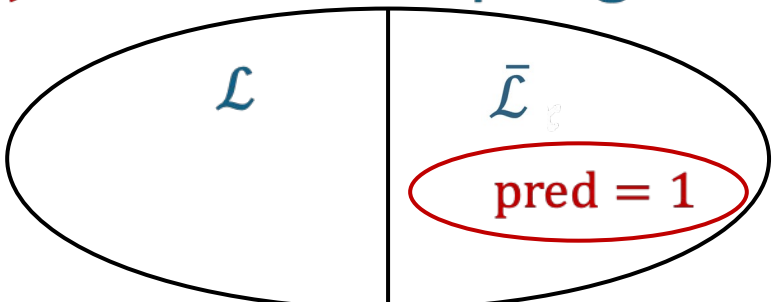
x, w



π

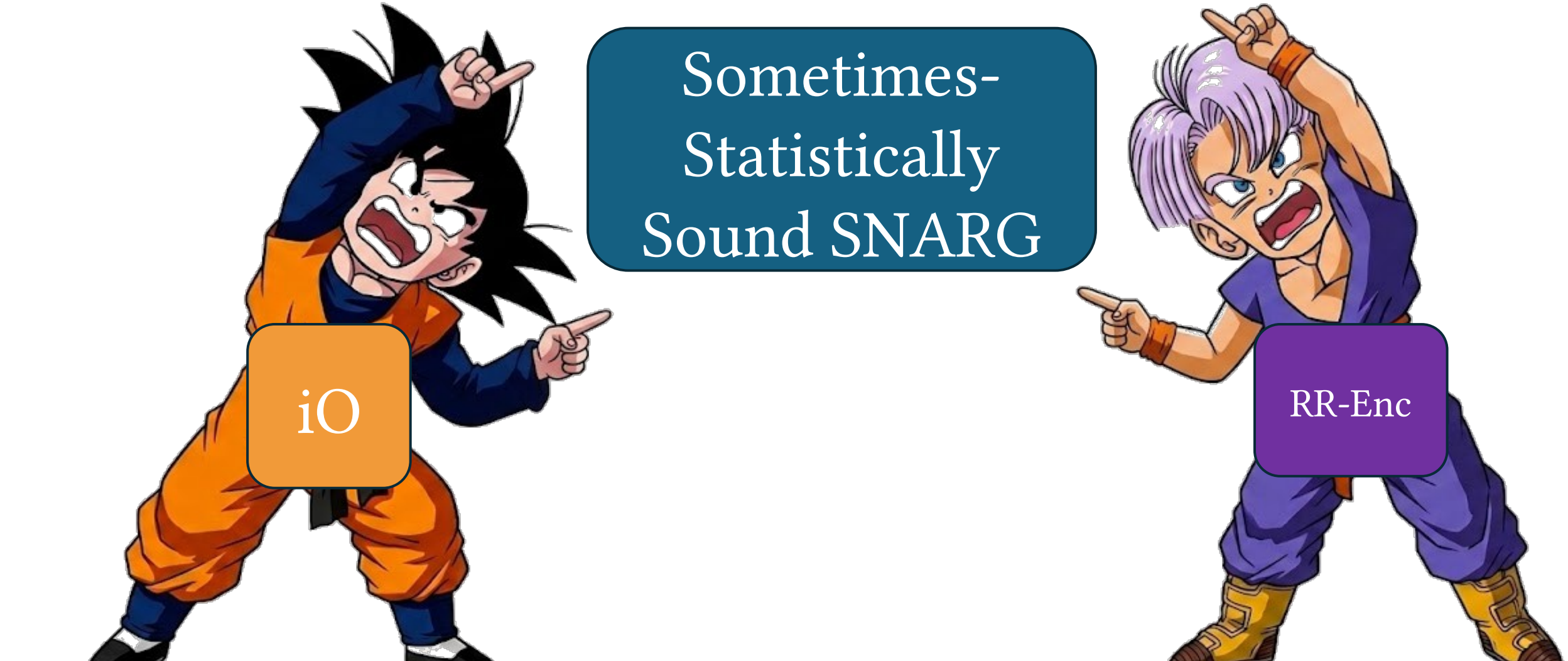
Statistically Sound: If $\text{pred}(x) = 1$, no accepting π exists

Mode Indistinguishability:
If

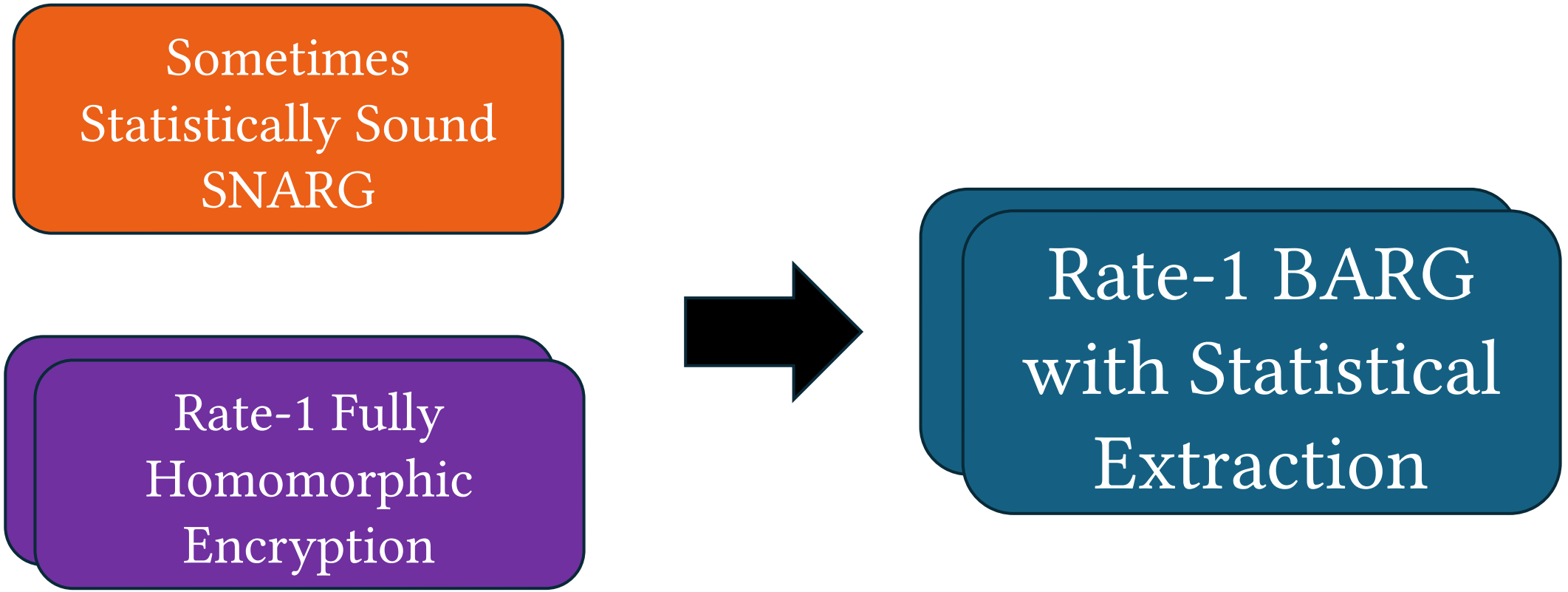


then CRS ≈ CRS′

This Work: Result 2



This Work: Result 3



Open Questions

- (Non-black box) construction of CRH from iO and plain OWFs?
- CRH from polynomially-secure iO and rerandomization?
- More applications of SSS-SNARGs



https://ia.cr/2026/1785